Graphs of Reciprocal Trigonometric Functions

Exam code: 9MA0

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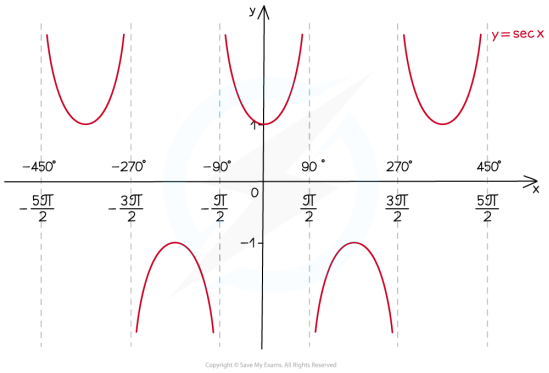
Help me understand

Graphs of reciprocal trigonometric functions

How do I sketch the graph of y = sec x?

- The graph of ***y* = sec*x*** looks like this:



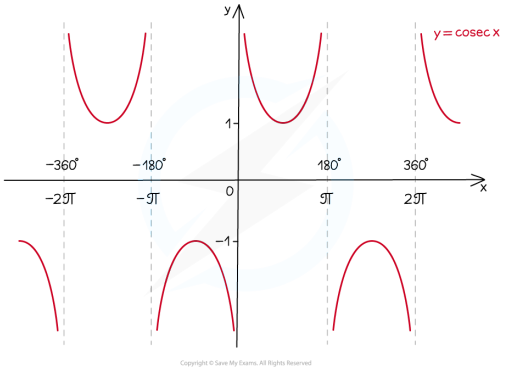
- ***y*-axis** is a **line of symmetry**
- has **period** (ie repeats every) **360°** or **2π** radians
- vertical **asymptotes** wherever **cos *x*= 0**

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

(π/2, - π/2, 3π/2, -3π/2, etc.)
- **range** is $y \leq -1$ **or** $y \geq 1$

How do I sketch the graph of y = cosec x?

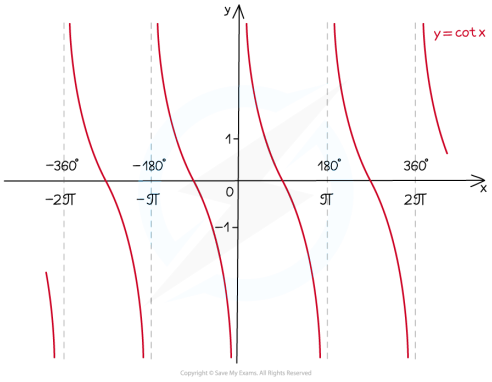
- The graph of $y = \text{cosec } x$ looks like this:



- has **period** (ie repeats every) **360°** or **2π** radians
- vertical **asymptotes** wherever **sin** $x = 0$
- **domain** is all x **except multiples of 180°** (0°, 180°, -180°, 360°, -360°, etc.)
- the **domain in radians** is all x **except multiples of π** (0, π, -π, 2π, -2π, etc.)
- **range** is $y \leq -1$ **or** $y \geq 1$

How do I sketch the graph of y = cot x?

- The graph of $y = \cot x$ looks like this:



- has **period** (ie repeats every) **180°** or **π** radians
- vertical **asymptotes** wherever **tan** $x = 0$
- **domain** is all x **except multiples of 180°** (0°, 180°, -180°, 360°, -360°, etc.)
- the **domain in radians** is all x **except multiples of π** (0, π, -π, 2π, -2π, etc.)

Examiner Tips and Tricks

- Make sure you know the shapes of the graphs for **cos**, **sin** and **tan**.
- The shapes of the reciprocal trig function graphs follow from those graphs plus the definitions **sec = 1/cos**, **cosec = 1/sin** and **cot = 1/tan**

Worked Example

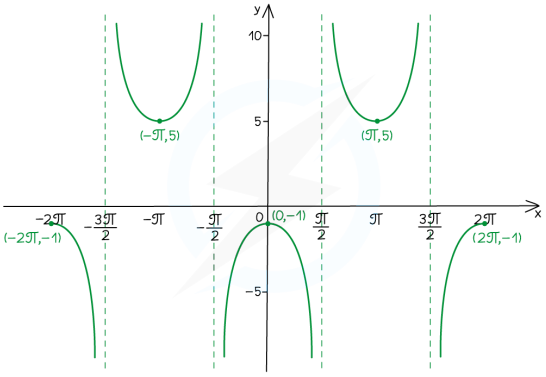
a) Sketch, in the interval $-2\pi \leq x \leq 2\pi$, the graph of $y = 2 - 3\sec x$.

b) Hence deduce the range of values of k for which the equation $2 - 3\sec x - k = 0$ has no solutions.

a)

THIS IS A TRANSFORMATION OF $y = \sec x$

- VERTICAL STRETCH, SCALE FACTOR 3
- THEN REFLECT IN x-AXIS
- FINALLY TRANSLATE VERTICALLY BY VECTOR $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$
- DON'T FORGET THE ASYMPTOTES



b) $2 - 3\sec x - k = 0$

$\Rightarrow 2 - 3\sec x = k$

FROM GRAPH, THE RANGE OF $y = 2 - 3\sec x$ IS $y \leq -1$ OR $y \geq 5$

SO THE EQUATION HAS NO SOLUTIONS WHEN $-1 < k < 5$

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